after the tear (using, say, the upper-right Calabi-Yau shape in Figure 11.5) and for its supposed mirror (the lower-right Calabi-Yau shape in Figure 11.5), and see if they are identical.

It was this calculation to which Aspinwall, Morrison, and I devoted ourselves in the fall of 1992.

Late Nights at Einstein's Final Stomping Ground

Edward Witten's razor-sharp intellect is clothed in a soft-spoken demeanor that often has a wry, almost ironic, edge. He is widely regarded as Einstein's successor in the role of the world's greatest living physicist. Some would go even further and describe him as the greatest physicist of all time. He has an insatiable appetite for cutting-edge physics problems and he wields tremendous influence in setting the direction of research in string theory.

The breadth and depth of Witten's productivity is legendary. His wife, Chiara Nappi, who is also a physicist at the Institute, paints a picture of Witten sitting at their kitchen table, mentally probing the edge of string theory knowledge, and only now and then returning to pick up pen and paper to verify an elusive detail or two.3 Another story is told by a postdoctoral fellow who, one summer, had an office next to Witten's. He describes the unsettling juxtaposition of laboriously struggling with complex string theory calculations at his desk while hearing the incessant rhythmic patter of Witten's keyboard, as paper after groundbreaking paper poured forth directly from mind to computer file.

A week or so after I arrived, Witten and I were chatting in the Institute's courtyard, and he asked about my research plans. I told him about the space-tearing flops and the strategy we were planning to pursue. He lit up upon hearing the ideas, but cautioned that he thought the calculations would be horrendously difficult. He also pointed out a potential weak link in the strategy I described, having to do with some work I had done a few years earlier with Vafa and Warner. The issue be raised turned out to be only tangential to our approach for understanding flops, but it started him thinking about what ultimately turned out to be related and complementary issues.

Aspinwall, Morrison, and I decided to split our calculation in two pieces. At first a natural division might have seemed to involve first extracting the physics associated with the final Calabi-Yau shape from